

# The Fermionic Matter Sub-Programme

## Presentation Note 6

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### Abstract

The fermionic matter sub-programme of the Cosmochrony corpus addresses a single central question: where do fermions, chirality, hypercharge, and three generations come from? The answer is that they are not postulated as external fields: they arise as the spinorial face of the admissible Weil module  $V_\rho$ , once its metaplectic Lie-algebraic structure is complexified by the Lorentzian metric established in the geometric branch. The structural chain is:

$$\Pi \Rightarrow F_n \simeq V_\rho \Rightarrow \mathrm{Mp}(2, \mathbb{R}) \Rightarrow \mathrm{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \rightsquigarrow \mathrm{Spin}(3, 1) \simeq \mathrm{SL}(2, \mathbb{C}) \Rightarrow \mathcal{S}_\Pi.$$

Three modular structural results are established in Q14: (A) the admissible spinor bundle  $\mathcal{S}_\Pi$  reproduces the  $\mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$  electroweak bundle structure from tensor functors of  $\mathcal{S}_\Pi$ ; (B) the projected Dirac operator  $D_{\Pi, g, A}$  contains a canonical projective endomorphism  $E_\Pi$  that enforces the  $V - A$  chiral structure (established via the spinorial BI lift theorem of Q14) and whose  $\gamma_5$ -weighted  $a_4$  coefficient imposes anomaly-cancellation constraints making hypercharge a spectral datum; (C) the saturation invariant  $\sigma_c(n_3) = 3$  has a spinorial multiplicity reading yielding a gauge-singlet three-generation factor  $C_{\mathrm{gen}}^3$ . The colour-coupled quark sector is unconditional at the pointwise level ( $[\mathrm{H}\text{-color}]_{\mathrm{pointwise}}$  established in O31). The structural form of  $E_\Pi$  is now fixed as a Schur complement of the eliminated spinorial block, and the generation-split amplitude mechanism is derived as Born–Infeld saturation in the companion note [9]; the open deliverables are its explicit Lorentzian block and the Yukawa sector, while the split value itself is dictionary-bound, its first-principles derivation relocated upstream to the ADE case selection and the cascade exponent.

**Keywords.** fermionic matter; admissible spinor bundle; metaplectic lift; chirality; V-A structure; hypercharge rigidity; anomaly cancellation; three generations; projected Dirac operator; Weil representation; non-injective projection; presentation note.

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# 1 Motivation and Central Question

The fermionic matter sub-programme answers the following question.

*Central question.* The bosonic spectral stratification — gravity at  $a_2$ , Yang–Mills at  $a_4$  — derives the bosonic sector of the Standard Model from the same admissible spectral functional. Fermions, chirality, hypercharge assignments, and the three-generation structure are not covered by the bosonic synthesis. *Do these fermionic structures also arise as forced consequences of the admissible Weil fibre, or must they be introduced as additional postulates?*

The answer established in Q14 [5] is: they are forced. Fermions arise as the spinorial face of the Weil module  $V_\rho$  after Lorentzian complexification of its metaplectic lift. Chirality and the  $V - A$  structure arise from the projective endomorphism  $E_\Pi$  of the projected Dirac operator, via the spinorial lift of BI parity established in Q14. Hypercharge is not a free parameter: it is constrained by spectral anomaly cancellation. Three generations arise from the spinorial multiplicity reading of the same rank-three invariant  $\sigma_c(n_3) = 3$  that yields the three spatial directions in the geometric branch.

This sub-programme is the first paper of the Q-series that depends simultaneously on all four layers of the programme: the admissible Weil fibre structure (Notes 1 and 5), the closed Lorentzian geometry (Note 2), the gauge bundle (Note 3), and the gauge–gravity synthesis (Q13 [6], to be summarised in Presentation Note 8).

**Remark 1** (Scope: identification only, not mass spectrum). This sub-programme closes the *identification* of the fermionic sector: spinor bundle, chirality, hypercharge quantum numbers, and generation factor. It does not derive the mass spectrum, the Yukawa couplings, or the CKM/PMNS mixing matrices. Those require the explicit construction of  $E_\Pi$  as an operator on the spectral space, and the integration of the projected Yukawa Lagrangian  $\mathcal{L}_Y$ , both of which are listed as open deliverables.

**Remark 2** (Vocabulary note). Throughout this note, *admissibility constraint* replaces the earlier “relaxation” vocabulary. The substrate  $\chi$  is static; fermions are not excitations of  $\chi$  but the spinorial interpretation of the admissible fibre after geometric complexification.

# 2 Position in the Cosmochrony Programme

The fermionic matter sub-programme sits at the apex of Branch III. It depends on all five preceding sub-programmes and provides the fermionic matter content consumed by any downstream phenomenological application.

Inputs from upstream:

- $F_n \simeq V_\rho \simeq L^2(\mathbb{Z}/q\mathbb{Z})$ , canonical symplectic structure of  $V_\rho$ , and the parity involution  $c \leftrightarrow q - c$  (Notes 1 and 5).
- Effective Lorentzian metric  $g^{\mu\nu} = 2\eta^{\mu\nu}$  and effective base manifold  $M_{\text{eff}} = \mathbb{R}_\tau \times \text{Heis}_3(\mathbb{R})$  (Note 2, Q5b–Q11).
- Admissible principal bundle  $P_{G_\Pi}(M_{\text{eff}}, G_\Pi)$  and gauge connection  $A$  (Note 3, Q6a).
- $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$  gauge group (Note 3).
- Spectral entropy functional  $S_\Pi[g, A]$  and the  $a_4$  variation (Note 4 [13]/Q12, Q13 [6]).
- $H_{\text{eff}} = \text{Sym}^2(V_\rho)$ , three spatial directions from  $\text{Im } \mathbb{H}$ ,  $\sigma_c(n_3) = 3$  (Note 2, Q7 [14]).

### 3 Logical Chain of the Sub-Programme

The sub-programme is organised around the following structural chain.

$$\begin{aligned} \Pi \Rightarrow F_n \simeq V_\rho \Rightarrow \text{Mp}(2, \mathbb{R}) \Rightarrow \text{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C}) \rightsquigarrow \text{Spin}(3, 1) \simeq \text{SL}(2, \mathbb{C}) \Rightarrow \mathcal{S}_\Pi \quad (1) \\ \Rightarrow \underbrace{\text{Sym}^2(\mathcal{S}_\Pi) \Rightarrow \text{ad}_{\mathbb{C}}(P_\Pi)}_{\text{SU}(2)_L}, \quad \underbrace{\wedge^2(\mathcal{S}_\Pi) = L_Y}_{\text{U}(1)_Y}, \quad \underbrace{E_\Pi \text{ left-admissible}}_{V-A}, \quad \underbrace{\sigma_c(n_3) = 3 \rightarrow C_{\text{gen}}^3}_{3 \text{ generations}}. \end{aligned}$$

Three distinct mechanisms contribute.

1. *Lorentzian complexification forces the spin group.* The Weil module  $V_\rho$  carries a canonical symplectic structure; its automorphisms lift to  $\text{Mp}(2, \mathbb{R})$ . The effective metric  $g^{\mu\nu} = 2\eta^{\mu\nu}$  (Lorentzian) forces the Lie algebra complexification  $\text{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$ , whose Lorentzian integration is  $\text{Spin}(3, 1) \simeq \text{SL}(2, \mathbb{C})$ . The spinor bundle  $\mathcal{S}_\Pi$  is the associated fundamental bundle.
2. *Tensor functors of  $\mathcal{S}_\Pi$  reproduce the electroweak bundle.*  $\text{Sym}^2(\mathbb{C}^2) \simeq \mathfrak{sl}_2(\mathbb{C})$ : the symmetric sector gives the complex adjoint, from which the compact real form selects the  $\text{SU}(2)_L$  gauge sector.  $\wedge^2(\mathbb{C}^2) \simeq \det(\mathbb{C}^2)$ : the determinant line  $L_Y$  supports the  $\text{U}(1)_Y$  factor. No independent electroweak matter bundle is introduced.
3. *The projective endomorphism  $E_\Pi$  enforces chirality and hypercharge.* The projected Dirac operator satisfies the Lichnerowicz-type identity

$$D_{\Pi, g, A}^2 = -(\nabla^{S, A})^2 + \frac{R}{4} + \frac{1}{2}\gamma^\mu \gamma^\nu F_{\mu\nu}^a T_R^a + E_\Pi.$$

By the spinorial lift of BI parity, established in Q14 [5],  $E_\Pi$  is left-admissible ( $P_R E_\Pi P_R = 0$ ), giving the  $V - A$  structure.  $\text{U}(1)_Y$  invariance of  $S_\Pi[g, A, D_{\Pi, g, A}]$  imposes the anomaly-cancellation trace constraint  $\text{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0$ , fixing hypercharge as a spectral coherence datum.

### 4 Constituent Papers

The sub-programme has a single primary paper.

#### 4.1 Q14: Fermionic matter and chirality from projective Dirac admissibility

**Q14** [5] is the sole constituent paper. It is organised around three independent but mutually compatible structural results.

*Theorem A: Spinorial electroweak bundle* (structural; unconditional). The admissible Weil module  $V_\rho$  induces, via the metaplectic lift and the Lorentzian complexification forced by  $g^{\mu\nu} = 2\eta^{\mu\nu}$ , an admissible spinor bundle  $\mathcal{S}_\Pi$ . Its tensor sectors satisfy:

$$\text{Sym}^2(\mathcal{S}_\Pi) \simeq \text{ad}_{\mathbb{C}}(P_\Pi), \quad \wedge^2(\mathcal{S}_\Pi) = L_Y.$$

After selection of the compact real form, these define the  $\text{SU}(2)_L \times \text{U}(1)_Y$  electroweak bundle data without any additional input.

*Theorem B: Chiral selection and hypercharge rigidity* (structural; unconditional). The projected Dirac operator  $D_{\Pi, g, A}$  has the Lichnerowicz-type square above. By the spinorial lift of BI parity, established in Q14 (the antilinear  $J_\Pi = HK$  maps right-chiral modes to non-admissible

directions),  $E_\Pi$  is left-admissible:  $P_R E_\Pi P_R = 0$ . This is the  $V - A$  structure.  $U(1)_Y$  invariance of the projected spectral functional then imposes, at the  $a_4$  level:

$$\text{Tr}_{S_\Pi}(\gamma_5 Y A_\Pi(x)) = 0 \quad \forall x \in M_{\text{eff}},$$

where  $A_\Pi$  is the  $\gamma_5$ -weighted  $a_4$  Seeley–DeWitt density of  $D_{\Pi,g,A}^2$ . This constraint is not postulated: it is the chiral consistency condition of the same  $a_4$  level that yields Yang–Mills dynamics in the bosonic variation. The hypercharge weights  $Y_R$  are thereby not free parameters but spectral coherence data.

*Theorem C: Generation multiplicity* (structural; unconditional). The saturation invariant  $\sigma_c(n_3) = 3$  has two functorially distinct readings of the same rank-three admissible invariant. In the geometric branch (Note 2), it yields three effective spatial directions. In the spinorial branch, the spinorial multiplicity functor maps it to a gauge-singlet generation space:

$$C_{\text{gen}}^3 \subset \ker(\text{ad}_{\text{SU}(2)} \oplus Y).$$

The identity  $\mathbb{R}_{\text{space}}^3 \neq C_{\text{gen}}^3$  is established explicitly (Q14 §5): the spatial and generation spaces are distinct functorial images of the same admissible invariant. Three generations are forced as a gauge-singlet factor. A qualitative mechanism for their splitting is identified in Q14 §6 (summarised below); the explicit  $E_\Pi$  and the amplitude of the splitting remain open.

*Dynamic generation lifting* (localisation and A4 matching). The internal structure of  $C_{\text{gen}}^3$  is probed by the spectrum of  $E_\Pi^2$  (Q14 §6). A static  $J_\Pi$ -real, weight-preserving restriction of  $E_\Pi^2$  has a degenerate outer pair: the two light generations cannot be split statically (static obstruction). The admissible lifts removing this degeneracy form a two-dimensional  $J_\Pi$ -odd sector, the real-split direction being the weight generator  $J_3$  and the mixing direction a doublet rotation. The generation split is not a finite-front observable in the symmetric sense: the raw finite front remains  $J_\Pi$ -equivariant and gives  $u_{\text{fin}} = 0$  [12]. After Lorentzian complexification, however, the chiral projector makes it meaningful to read a  $J_\Pi$ -odd frontier cocycle, and the residual automorphism is the central Weil lift  $R_b = W(-I) = \text{FT}^2$ , which commutes with  $\gamma_5$ : it is blind to chirality and does not obstruct the  $V - A$  selection [11]. The remaining amplitude is fixed at A4 by matching the discrete chiral-frontier normalisation  $\mathcal{N}_A$  [7] with the Lorentzian Born–Infeld curvature  $\mu_\chi^2 = \partial_s^2 \mathcal{B}(0)$  [12], neither ingredient reducing to the other. Generation mixing further requires the complex metaplectic phase.

*Colour sector* (unconditional at the pointwise level). The colour-coupled quark sector  $S_\Pi \otimes V_{\text{color}}$  is obtained by tensoring the admissible spinor bundle with the colour module  $V_{\text{color}}$ . The status of this sector inherits the status of [H-color] from Notes 1 and 3: established unconditionally on the colour-adapted graph, and on the standard graph at the rank, averaged, and effective-exponent levels, with [H-color]<sub>pointwise</sub> proved analytically (O31, exact equality of BFS capacity profiles at finite  $q$ ). The only residual component is the extension beyond the standard Cayley graph to general generating sets, which is non-blocking for the quark bundle.

*Status of the spinorial BI lift.* The spinorial lift of BI parity to the spinor bundle  $S_\Pi$  is established as a theorem in Q14: the BI parity involution  $c \leftrightarrow q - c$ , proved at the scalar Weil level in O18 and BornInfeld, lifts to the antilinear endomorphism  $J_\Pi = HK$  satisfying  $J_\Pi^2 = -1$  and  $J_\Pi \gamma_5 = -\gamma_5 J_\Pi$ , dressing the parity by the weight grading  $H = 2J_3$ . Chirality and  $V - A$  in Q14 are therefore unconditional, their only residual dependence being the axiomatic chain A1–A4 [1] and the upstream BI parity datum (O30, U1).

## 4.2 Angular Amplitude Reduction: the $J_3$ split as oriented symplectic area [7]

This note fixes the numerical observable for the absolute normalisation of the angular generation split before any Weil–BFS computation. A Baker–Campbell–Hausdorff reduction identifies the per-step  $J_\Pi$ -odd  $J_3$  component of the metaplectic cascade with the oriented symplectic area swept per step — the central, Carnot-degree-2 increment of  $\text{Heis}_3$  — and not with a free shear

Table 1: Summary of Q14 results by theorem. Status: P = proved; S = structural.

| Result                  | Central output                                                   | Status                 |
|-------------------------|------------------------------------------------------------------|------------------------|
| Theorem A               | Spinor bundle $\mathcal{S}_\Pi$ ; EW bundle from tensor functors | S (uncond.)            |
| Theorem B ( $V - A$ )   | $P_R E_\Pi P_R = 0$ ; $V - A$ structure                          | S (uncond.)            |
| Theorem B (hypercharge) | $Y_R$ from anomaly-cancellation trace; not free                  | S (uncond.)            |
| Theorem C               | $C_{\text{gen}}^3$ ; three generations as gauge-singlet factor   | S (uncond.)            |
| Colour sector           | $\mathcal{S}_\Pi \otimes V_{\text{color}}$ ; quark bundle        | S (uncond., pointwise) |

parameter. The quantity to measure is the accumulated oriented  $J_3$ -odd area along the cascade, normalised by the projected capacity. A direct evaluation on symmetric BFS shells returns  $\Theta_{\text{raw}} = 0$  identically: the symmetric capacity data fix the radial sector but cannot select an oriented angular branch, so the orientation prescription is an irreducible input prior to the normalisation  $\mathcal{N}_A$ .

### 4.3 Q11 Oriented Frontier: recursive diagnostic and resolution of the operator obstruction [11]

The symmetric-shell observable vanishes ( $\Theta_{\text{raw}} = 0$ ) because the involution  $g \mapsto g^{-1}$  pairs opposite central phases; this note defines the honest replacement, a frontier transfer observable on the directed edges  $g \rightarrow gs$  whose central increment is the Heisenberg cocycle  $a(g)s_b$ , the discrete counterpart of  $\frac{1}{2}[X, Y]$  and hence of the dynamical  $J_3$ . Its main result resolves the operator obstruction: the residual reflection  $R_b = \varphi \circ J_\Pi^{-1} = W(-I) = \text{FT}^2$  is the central Weil lift, and  $[\gamma_5, R_b] = 0$ , so  $R_b$  is blind to chirality and the  $V - A$  selector removes the  $\varphi$ -paired obstruction without a spacetime parity. What remains is the amplitude normalisation  $\mathcal{N}_A$ , not the operator question. The chiral weight  $\gamma_5$  entering this signal is identified in CHO-Note [4] as the Weyl representation type **2** versus  $\bar{\mathbf{2}}$ , not a function of the cascade generator  $T(e)$  (arrow inversion and  $\varphi$  both send  $T \mapsto -T$ , fixing equal parities, while the admissible weight is arrow-even and  $\varphi$ -odd); in the inherited Heisenberg central-phase lift the outgoing-frontier overlap is maximal,  $\rho_\chi = 1$ , so  $\mathcal{N}_A = \theta_{\text{max}} \neq 0$  in that lift.

### 4.4 Projective Residue as a Schur Complement: localisation and finite/Lorentzian separation [12]

This note fixes the structural status of the split  $u$  before the explicit  $E_\Pi$ . The projected Dirac square has the Schur–Feshbach form  $E_\Pi = -M^\dagger M$  with  $M = (1 - P) D \Pi_S^*$ , exhibiting  $E_\Pi$  as the Schur complement of the eliminated spinorial block;  $u$  is localised in the  $J_\Pi$ -odd part of  $E_\Pi^2|_{C_{\text{gen}}^3}$  as the antiunitary equivariance defect  $\Delta_\chi(P) = \pi_{LL} - \tau \bar{\pi}_{RR} \tau^{-1}$ . The finite locking is  $J_\Pi$ -equivariant, so  $u_{\text{fin}} = 0$  (finite/Lorentzian separation): the split cannot originate in the finite cascade and is a Lorentzian datum, its magnitude reduced to the Born–Infeld saturation curvature  $\mu_\chi^2 = \partial_s^2 \mathcal{B}(0)$ . The Schur transversality on which the A4 matching was conditional is closed in the present Lorentzian spin stratum [12]: the projected commutator  $[\partial_s P|_0, J_\Pi]|_{\text{span}(e_+, e_-)}$  has the exact metaplectic opening  $\alpha(t, s) = tsr / \sinh r \neq 0$  with  $\mu(t, s) \equiv 0$ , and its timelike Clifford symbol is not in the zero-mode locus, so the Schur-transverse branch is selected. The determinantal Born–Infeld margin  $\mathcal{B}(s)$ , the antisymmetric structure of the chiral modulus, and the reduction of  $\mu_\chi^2$  to the Lorentzian genus of the chiral polarisation are developed in the companion note [9]. That companion further derives the amplitude mechanism: on the corpus-derived real symplectic cascade the transverse (spin-two) cubic response vanishes path-invariantly at the complex metaplectic phase  $\gamma = 0$ , for every ordering and reference, so the derived mechanism is Born–Infeld saturation while the split persists ( $u \neq 0$ ). The sixth-order interior lock is dou-

bly conditional — on a non-derived complex phase and on a non-prescribed cascade ordering — and not concluded; the split value  $|u|$  remains dictionary-bound through the chiral-frontier normalisation  $\mathcal{N}_A$ .

#### 4.5 Arithmetic orthogonality of the ADE gate: the conditional status of $\varepsilon$ [8]

This transversal note isolates the logical status of the generation-split value  $\varepsilon = 1/10$ . It is a theorem modulo the ADE case-selection gate: the algebra  $\frac{1/2+\varepsilon}{1/2-\varepsilon} = \frac{3}{2} \Rightarrow \varepsilon = 1/10$  is trivial and case-independent, so the first-principles status of  $\varepsilon$  is exactly that of the gate. The obstruction to the gate is characterised as an arithmetic orthogonality. In the compositum  $\mathbb{Q}(\zeta_q, \sqrt{5})$  the Born–Infeld parity  $c \mapsto q - c$  is complex conjugation ( $\zeta_q \mapsto \zeta_q^{-1}$ ,  $\sqrt{5} \mapsto \sqrt{5}$ ), lying in a distinct direct factor of the Galois group from the Galois-spin action ( $\zeta_q \mapsto \zeta_q$ ,  $\sqrt{5} \mapsto -\sqrt{5}$ ) that selection of the  $\sqrt{5}$ -locus would require. The same fact places the order-five versus order-ten alternative below projective parity resolution, consistent with the central reflection  $R_b = W(-I)$  being scalar on the chiral carrier [11]. The negative theorem — that A1–A4, ENI Corollary 6 and the A4 locking do not supply the Galois-spin action — is a no-go for the gate, unconditional in the present corpus construction by a character-table audit of  $2I$ ; only the full-tower generalisation to every conceivable recursive stratum remains open. Hence  $\varepsilon = 1/10$  is conditional on the gate rather than an independent dictionary parameter, and the first-principles front for the value is the gate itself: upstream and structural.

#### 4.6 Chiral orientation of the directed frontier: $\sigma_L$ as a representation type [4]

This note resolves the chiral weight  $\sigma_L$  left deferred by Q11OF. A parity argument closes the false route  $\sigma_L = f(T(e))$ : edge inversion  $U(e^{-1}) = U(e)^{-1}$  and the residual reflection both send the  $J_{\Pi}$ -odd cascade generator  $T(e) \mapsto -T(e)$ , so every function of  $T$  has equal arrow- and  $\varphi$ -parity, whereas the admissible weight must be arrow-even and  $\varphi$ -odd. The resolution is that  $\sigma_L$  is the Weyl representation type **2** versus  $\bar{\mathbf{2}}$ : group inversion preserves the representation (arrow-even) and complex conjugation exchanges it ( $\varphi$ -odd), the unique label distinguished by the conjugation that separates  $\varphi$  from arrow inversion; for a real lift the representation is self-conjugate and the contrast vanishes, matching the chiral symmetry of the real cascade. In the inherited Heisenberg central-phase lift the edge phase is the linear cyclotomic character  $\zeta_q^{\Delta A_c}$ , with no Gauss, Maslov, or  $\sqrt{5}$  component; the natural outgoing branch gives  $\sigma_L = \text{sign } \Delta A_c$  on the cocycle support, hence  $\rho_\chi = 1$  and  $\mathcal{N}_A = \theta_{\max} \neq 0$ ,  $q$ -invariant on  $q \in \{61, 101, 151\}$ . The result is therefore [H-orient] closed in the inherited central-phase lift; the residual open point is an additional spin-Galois phase of the full Lorentzian chiral lift, in particular a  $\sqrt{5}$  or ADE-spin factor orthogonal to  $\zeta_q$ , the same arithmetic orthogonality that governs the ADE gate [8]. It fixes the frontier amplitude datum conditionally and does not derive  $u \neq 0$ , which still requires the Schur transport [12].

### 5 Inputs from Upstream Results

1.  **$F_n \simeq V_\rho$  and symplectic structure** (Notes 1, 5): the identification of the admissible fibre as the Weil representation is the starting point of the metaplectic lift. Without this, the spinor bundle  $\mathcal{S}_{\Pi}$  has no canonical definition.
2. **Lorentzian metric  $g^{\mu\nu} = 2\eta^{\mu\nu}$**  (Note 2, Q5b–Q11): the Lorentzian signature forces the complexification  $\text{mp}(2, \mathbb{R})_{\mathbb{C}} \simeq \mathfrak{sl}_2(\mathbb{C})$ . Without the Lorentzian metric, the spin group would remain  $\text{Sp}(2, \mathbb{R})$  and no Dirac spinors would arise.
3. **Gauge bundle and  $\text{SU}(2)_L \times \text{U}(1)_Y \times \text{SU}(3)$**  (Note 3, Q6a): the admissible principal bundle and gauge connection  $A$  enter the spin–gauge connection  $\nabla^{S,A}$  of the projected Dirac operator.

4.  **$a_4$  Yang–Mills variation** (Q12 [15], Q13 [6]): the spectral anomaly cancellation of Theorem B uses the same  $a_4$ -level structure that yields Yang–Mills dynamics in the bosonic sector. The fermionic  $a_4$  coefficient is a structural extension of the bosonic one. The gauge–gravity synthesis of Q13 is the scientific dependency here; its presentation-level treatment will appear in Presentation Note 8.
5.  $\sigma_c(n_3) = 3$  (Note 1, O23): the rank-three saturation invariant provides the spinorial multiplicity input for Theorem C.
6. **BI parity involution** (BornInfeld, O18): the scalar BI parity  $c \leftrightarrow q - c$  is the input from which the spinorial BI lift is derived; its derivation at the scalar level is complete.

## 6 Outputs to Downstream Branches

Table 2: Principal outputs and downstream consumers.

| Output                                                  | Consumer                | Status                 |
|---------------------------------------------------------|-------------------------|------------------------|
| Admissible spinor bundle $\mathcal{S}_\Pi$              | Phenomenology, Q-series | S                      |
| $SU(2)_L \times U(1)_Y$ from tensor functors            | SM phenomenology        | S                      |
| $V - A$ chiral structure                                | Weak interaction pheno  | S (uncond.)            |
| Hypercharge as spectral datum                           | SM charge assignment    | S (uncond.)            |
| Three-generation factor $C_{\text{gen}}^3$              | Mass/mixing notes       | S                      |
| Quark bundle $\mathcal{S}_\Pi \otimes V_{\text{color}}$ | QCD sector              | S (uncond., pointwise) |
| $E_\Pi$ as projective endomorphism                      | Yukawa sector (open)    | S (open)               |

## 7 Status of Results

### 7.1 Structural results (unconditional)

1. *Admissible spinor bundle* (Theorem A):  $\mathcal{S}_\Pi$  is induced from  $V_\rho$  via the metaplectic lift and Lorentzian complexification. The construction is algebraic and functorial; it depends on the Lorentzian closure of Q5b–Q11 but not on the spinorial BI lift.
2.  $SU(2)_L$  from  $\text{Sym}^2(\mathcal{S}_\Pi)$  (Theorem A): the symmetric tensor sector gives the complex adjoint  $\text{ad}_{\mathbb{C}}(P_\Pi) \simeq \mathfrak{sl}_2(\mathbb{C})$ ; after compact real-form selection, the  $SU(2)_L$  gauge sector is identified.
3.  $U(1)_Y$  from  $\wedge^2(\mathcal{S}_\Pi)$  (Theorem A): the determinant line  $L_Y = \wedge^2(\mathcal{S}_\Pi)$  is the geometric support of the hypercharge factor.
4. *Generation multiplicity* (Theorem C):  $C_{\text{gen}}^3 \subset \ker(\text{ad}_{SU(2)} \oplus Y)$  is a gauge-singlet three-generation space arising from the spinorial multiplicity reading of  $\sigma_c(n_3) = 3$ . The spatial and generation readings of the rank-three invariant are distinct ( $\mathbb{R}_{\text{space}}^3 \neq C_{\text{gen}}^3$ ; Q14 §5).

### 7.2 Further structural results (unconditional)

1. *Left-admissibility of  $E_\Pi$*  (Theorem B):  $P_R E_\Pi P_R = 0$  follows from the spinorial BI lift (Q14), now established.



2. *V – A chiral structure* (Theorem B): the condition  $P_R E_\Pi P_R = 0$  selects the left-chiral admissible projection  $P_L \mathcal{S}_\Pi$ , giving the  $V – A$  coupling of weak interactions.
3. *Hypercharge rigidity* (Theorem B): the anomaly-cancellation trace condition  $\text{Tr}_{\mathcal{S}_\Pi}(\gamma_5 Y A_\Pi(x)) = 0$  constrains  $Y_R$  to spectral coherence data; hypercharge is not a free parameter of the framework.
4. *Quark bundle* (§6):  $\mathcal{S}_\Pi \otimes V_{\text{color}}$  is structural and unconditional at the pointwise level: the spinorial BI lift is established (Q14) and  $[\text{H-color}]_{\text{pointwise}}$  is proved analytically (O31).
5. *A4 Schur-transversality closure* [12]: the conditional residual of the A4 matching is closed in the present Lorentzian spin stratum — the projected commutator  $[\partial_s P|_0, J_\Pi]_{\text{span}(e_+, e_-)}$  has the exact metaplectic opening  $\alpha(t, s) = tsr/\sinh r \not\equiv 0$ ,  $\mu(t, s) \equiv 0$ , and its timelike Clifford symbol is not in the zero-mode locus, so the Schur-transverse branch is selected and the split is determined via the electric genus of the chiral polarisation [9].

### 7.3 Open problems

1. *Explicit form of  $E_\Pi$* .  $E_\Pi$  is identified as the zero-order endomorphism of the projected Dirac square, the “internal spectral residue of non-injective projection.” Its structural form is now fixed as the Schur complement of the eliminated spinorial block,  $E_\Pi = -M^\dagger M$  with  $M = (1 - P)D\Pi_S^*$  [12]; what is not yet constructed is its explicit Lorentzian block  $1 - P(s)$  along the  $J_\Pi$ -odd modulus. The inter-generation splitting is localised in the  $J_\Pi$ -odd part of  $E_\Pi^2$ ; its amplitude mechanism is the derived Born–Infeld saturation [9]. The split value  $\varepsilon = 1/10$  is not an independent dictionary parameter: it is forced algebraically once the ADE case-selection gate supplies the external 3 : 2 ratio. Its first-principles status is therefore exactly the status of that gate, whose obstruction is isolated in AOG-Note [8] as an arithmetic orthogonality between Born–Infeld parity and the Galois-spin action. Without the explicit Lorentzian  $E_\Pi$ , the three-generation mass spectrum and the Yukawa sector cannot be derived.
2. *Normalisation of  $\mathcal{L}_Y$* . The Yukawa Lagrangian  $\mathcal{L}_Y$  is identified as arising from the trace  $\text{Tr}_{\mathcal{S}_\Pi}[E_\Pi^2]$  variation. Its integral normalisation and the derivation of the mass matrix from the spectral data of  $E_\Pi$  are open.
3.  $[\text{H-color}]_{\text{pointwise}}$  *beyond the standard graph*. Inherited from Notes 1 and 3.  $[\text{H-color}]_{\text{pointwise}}$  is proved on the standard Cayley graph (O31); its extension to general generating sets is the only residual, and is non-blocking for the quark bundle, which is already established at the pointwise level.

## 8 Open Deliverables

The derivation of the spinorial BI lift, formerly the first open deliverable, is now established as a theorem in Q14 [5]: the antilinear lift  $J_\Pi = HK$  dresses the BI parity by the weight grading, making Theorems B and C unconditional. Two open deliverables remain.

**Open deliverable 1: Explicit  $E_\Pi$  and the Yukawa sector.** The mass spectrum of charged leptons and quarks, and the CKM/PMNS mixing matrices, require the explicit spectral construction of  $E_\Pi$  as an operator on the admissible Hilbert space. This requires integrating the projected Yukawa Lagrangian  $\mathcal{L}_Y$  and extracting the mass matrix from the eigenvalues of  $E_\Pi^2$ . This is the primary open problem of the fermionic sector. The structural form of  $E_\Pi$  is now fixed as the Schur complement of the eliminated spinorial block [12]; the open part is its explicit Lorentzian block  $1 - P(s)$  along the  $J_\Pi$ -odd modulus. The symmetric finite front is  $J_\Pi$ -equivariant and gives  $u_{\text{fin}} = 0$ , but the discrete chiral-frontier normalisation  $\mathcal{N}_A$  survives the

operator obstruction, since the residual reflection  $R_b = \text{FT}^2$  is blind to chirality ( $[\gamma_5, R_b] = 0$ ) [11], and is fixed maximally,  $\mathcal{N}_A = \theta_{\max}$ , in the inherited central-phase lift [4]. The split amplitude is therefore fixed at A4 by matching this discrete normalisation with the Lorentzian Born–Infeld curvature  $\mu_\chi^2 = \partial_s^2 \mathcal{B}(0)$ , neither ingredient reducing to the other:

$$|u| = \text{A4Match}(\mathcal{N}_A, \mu_\chi^2).$$

The operator side of this matching is no longer conditional: by [12] the Schur transversality is closed in the present Lorentzian spin stratum, the projected commutator  $[\partial_s P|_0, J_\Pi]_{\text{span}(e_+, e_-)}$  having the exact metaplectic opening  $\alpha(t, s) = tsr/\sinh r \not\equiv 0$  with timelike Clifford symbol off the zero-mode locus. In the Schur-transverse branch the A4 chiral polarisation is electric, hence  $\mathcal{P}_{\mu\nu}\mathcal{P}^{\mu\nu} < 0$ ,  $\mu_\chi^2 < 0$ , and the split opens spontaneously [9]; the genus is determined. What remains open is the split value  $|u|$  and the Yukawa sector downstream of  $u$ . On the corpus-derived real cascade the A4 companion resolves the boundary-versus-interior selection in favour of saturation: the transverse (spin-two) cubic response of  $\mathcal{P} \cdot \mathcal{R}_\chi$  vanishes path-invariantly at  $\gamma = 0$ , so the derived amplitude mechanism is Born–Infeld saturation, and the sixth-order interior lock is doubly conditional — on a non-derived complex phase and a non-prescribed ordering — hence not concluded [9]. The value  $|u|$  itself is not a parameter-free prediction: it is dictionary-bound through the chiral-frontier normalisation  $\mathcal{N}_A$ , the angular analogue of the radial scale  $\kappa = 5/12$  of the exit-deficit dictionary [7, 5], whose split value  $\varepsilon = 1/10$  is fixed by matching the derived three-level deficit structure to the external ADE ratio of a selected case [3]. The genuine first-principles front for the value is therefore upstream and structural, not a numerical amplitude: the ADE case selection and the level-to-generation map, the derivation of the projective-resolution growth and the cascade exponent  $\beta$  (only  $\beta \leq 1$  established [3, 2]), and the derivation of the transfer constant  $N_{\text{casc}}$  that fixes  $\mathcal{N}_A$  [5]. What remains quantitative is therefore these upstream dictionary inputs, the integral normalisation of  $\mathcal{L}_Y$ , and generation mixing via the complex metaplectic phase.

**Open deliverable 2: Full matter content in Lorentzian signature.** The complete fermionic content  $S_\Pi^{\text{matter}} = (\mathcal{S}_\Pi \oplus (\mathcal{S}_\Pi \otimes V_{\text{color}})) \otimes C_{\text{gen}}^3$  in full Lorentzian signature, with explicit gauge couplings and three-generation Yukawa structure, is the downstream target. It requires the resolution of Deliverable 1 above, the colour sector being already established at the pointwise level (O31).

## 9 Conclusion

The fermionic matter sub-programme establishes that fermions, chirality, hypercharge, and three generations are not external additions to the Cosmochrony framework. They are the spinorial face of the admissible Weil module  $V_\rho$ , revealed through the Lorentzian complexification of its metaplectic structure.

The electroweak bundle  $SU(2)_L \times U(1)_Y$  arises from the tensor functors of the admissible spinor bundle  $\mathcal{S}_\Pi$  (Theorem A, unconditional). The  $V - A$  chiral structure and hypercharge rigidity arise from the projective endomorphism  $E_\Pi$  (Theorem B, unconditional). Three generations arise from the spinorial multiplicity reading of  $\sigma_c(n_3) = 3$  (Theorem C, unconditional). The colour-coupled quark sector follows by tensoring with  $V_{\text{color}}$ , unconditional at the pointwise level (O31).

The inter-generation splitting is localised: statically obstructed, living in a two-dimensional  $J_\Pi$ -odd lift sector, and vanishing on the symmetric finite front ( $u_{\text{fin}} = 0$ ,  $J_\Pi$ -equivariant). Its operator obstruction is resolved, the residual reflection  $R_b = \text{FT}^2$  being blind to chirality ( $[\gamma_5, R_b] = 0$ ), and its amplitude mechanism is the derived Born–Infeld saturation [9]. The remaining open components are the explicit Lorentzian  $E_\Pi$ , whose eliminated-block jet  $1 - P(s)$  is structurally constrained and whose chiral generator is identified as the transported equivariance defect rate  $[\Pi_{J_\Pi\text{-odd}}\dot{Q}(0)]_{LL} = \frac{1}{2}\partial_s\Delta_\chi(P)|_0$  in the companion note [10] (reducing the split value to the single normalisation target  $\partial_s\Delta_\chi(P)|_0$ ); the split value, which is dictionary-bound and whose first-principles derivation is upstream and structural (the ADE case selection and level-to-generation map, the cascade exponent  $\beta$ , and the transfer constant  $N_{\text{casc}}$ ); and the derivation of the Yukawa sector and mass spectrum.

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